Predicting Next Day Step Counts Using Fitbit Activity Records

1 A Forecasting Question for the Prototype

Wearable activity records offer an accessible setting for studying short-term behavioral patterns. This project will build a small prediction tool for estimating a user's next-day step count from recent steps, activity intensity and calendar context.

The research question is whether short-term trends and activity composition improve forecasting beyond simple personal baselines, and whether any gain holds when activity changes sharply. The project emphasizes a working prototype and a controlled comparison, rather than a new algorithm or a clinical intervention.

2 The Forecasting Decision

Seven-day trends and activity intensity may reduce next-day prediction error beyond today's count or a rolling-mean forecast. Extra features may also overfit this small dataset, so the project will test their incremental value rather than assume that a larger input set is better.

2.1 Four Ways to Predict Tomorrow

1. Implement four predictors in Python: persistence, a seven-day rolling mean, ridge regression and a small random forest regressor. Persistence repeats today's steps; the rolling mean represents a person's recent routine. The learned models will be compared under three nested feature sets on the same eligible days.
2. Workflow: validated daily records → chronological feature construction → baseline and model predictions → comparison with observed next-day steps → result display.

   A simple FastAPI endpoint and lightweight page will display the recent activity trajectory, a model prediction, the rolling-mean baseline and the observed value when available. Pydantic will validate inputs. The tool is educational; it will not generate medical advice or exercise prescriptions.

2.2 A Meaningful Difference in Error

Use mean absolute error (MAE), measured in steps, as the primary metric and root mean squared error (RMSE) as a secondary metric. Report each model on identical test dates, alongside participant and observation counts. Also show the median participant-level MAE and descriptive errors for weekdays versus weekends and

stable versus volatile recent activity; small subgroups will not support significance claims.

2.3 Why These Methods

The Fitbit dataset provides real-world wearable records for exploratory activity analysis (arashnic, n.d.). Its daily activity fields include steps, activity-intensity minutes, sedentary minutes and estimated calories (Fitabase, n.d.). Random forests are an established structured-data comparator to a linear model (Breiman, 2001).

This project applies existing methods to a narrow forecasting task. Its added analysis is a controlled comparison of feature groups and errors across different activity patterns, not methodological novelty. Independent validation would still be needed for stronger scientific conclusions.

3 Turning Fitbit Records into Examples

We will use the Fitbit Fitness Tracker Dataset explicitly listed under Wearable Device Datasets . The designated source is https://www.kaggle.com/datasets/arashnic/fitbit. The daily activity table supplies all required fields; no additional dataset is needed.

- Predictors: recent steps and target weekday; seven-day step trend and variability; and lagged activity-intensity, sedentary-time and calorie measures.
- Target: the recorded total number of steps on the next calendar day for the same participant.

The repository describes a small volunteer sample with activity, heart-rate and sleep records. This project uses only the daily activity table, avoiding a sparse cross-file join. Usable participant counts and seven-day windows will be reported after cleaning rather than assumed from the repository description.

3.1 Dates Inputs and Cutoffs

1. Check participant identifiers, dates, missing values, duplicate participant-day rows and implausible daily totals. Retain genuinely consecutive seven-day windows; a later observed record must not stand in for the next calendar day when dates are missing. Zero steps are reviewed but not automatically treated as missing.
2. From days t-6 through t, calculate three- and seven-day step means, seven-day step standard deviation and the latest day-to-day change. Add the known weekday of t+1 and day-t minutes spent lightly, fairly and very active, sedentary minutes and estimated calories. Use participant identifiers only to group records; every input must be available by the end of day t.
3. Split observations by target date into approximately 60% training, 20% validation and 20% testing, with common chronological cutoffs across participants. Fit scaling and any

> imputation on training data only; select settings on validation data and freeze choices before testing. Report actual split sizes and exclude test participants absent from training.

4 Running the Forward Test

This is a supervised regression task using a single source of wearable data. Forecasts are generated at the end of day t for day t+1, using only information available through day t. The main analysis concerns future observations from participants already represented in the training period.

4.1 One Comparison on Later Days

Evaluate both simple baselines and the two learned models on the same held-out seven-day-eligible observations. For each learned model compare three nested feature sets: A, recent steps plus weekday; B, A plus seven-day trend and variability; C, B plus lagged activity-intensity minutes, sedentary minutes and estimated calories. Use a small validation search for the ridge penalty and forest depth, with a fixed random seed.

> The feature-block comparison will show whether added signals justify their data and engineering cost. Deep learning, extra sensor files, broad hyperparameter searches and external validation remain outside scope.

4.2 Looking Beyond the Average Error

Plot predicted versus observed steps and activity traces for several participants. Compare errors for target weekdays and weekends and for low- versus high-variability seven-day windows. Review selected large misses to see whether sharp routine changes defeat all methods or mainly the simpler baselines.

> These are descriptive checks, not causal explanations of activity changes. Recorded steps, activity minutes and estimated calories are device-derived measures rather than direct measures of overall health.

5 Limits and Handoff

1. Data quality: zero steps can reflect inactivity or incomplete device wear, while activity minutes and calories may be missing or inconsistent. Document cleaning decisions and feature availability; do not infer non-wear reliably from a zero alone.
2. Limited predictability and overfitting: schedules and personal events can change activity abruptly, while the expanded feature set is large relative to this volunteer sample. Keep tuning small and report a negative result if richer models fail to beat simple baselines.

3. Generalizability: a small volunteer sample and a short observation period restrict conclusions. The chronological evaluation does not demonstrate performance for new users, other devices or clinical populations.
4. Research scope: the project does not test interventions, demographic fairness or clinical outcomes. Its contribution is an initial feature-group comparison, time-aware evaluation and a basic data application.

5.1 Making the Work Reproducible

Host the code on GitHub with the group number and student names. Provide dependencies, the feature recipe, data-cleaning rules, date cutoffs, fixed seed, model settings and evaluation commands. Share access with the teaching staff before the report deadline.

Sources

1. arashnic. (n.d.). FitBit fitness tracker data [Data set]. Kaggle. https://www.kaggle.com/datasets/arashnic/fitbit
2. Breiman, L. (2001). Random forests. Machine Learning, 45, 5–32. https://doi.org/10.1023/A:1010933404324
3. Fitabase. (n.d.). Fitbit data dictionary. https://www.fitabase.com/media/2101/fitabase-fitbit-data-dictionary-as-of-11325.pdf